2025학년도 왕운중3 1학기 중간 수학

1. $\boxed{2번}$

① a의 제곱근은 $\pm\sqrt{a}$

② 양수는 2개, 0은 1개
음수는 없다

④ -16의 제곱근은 없다

⑤ 제곱근 81은 9

2. $\boxed{3번}$

$\sqrt{(-16)^2} = |-16| = 16$

의 양의 제곱근 $a=4$

제곱근 81은 $\sqrt{81} = 9$

의 음의 제곱근 $-3=b$

$ab = -12$

3. $\boxed{1번}$

$\frac{\sqrt{15}}{3} \times \frac{\pi}{\sqrt{5}} \div \frac{\pi}{\sqrt{12}}$

$= \frac{\sqrt{15}}{3} \times \frac{\pi}{\sqrt{5}} \times \frac{\sqrt{12}}{\pi}$

$= 2$

4. $\boxed{3번}$

순환소수가 아닌 무한소수 = 무리수

π^2, $\sqrt{3.\dot{7}}$, $\sqrt{8.1}$

5. $\boxed{2번}$

① $3 > \sqrt{2}+1$

② $\sqrt{5}-\sqrt{2} < \sqrt{7}-\sqrt{2}$

③ $\sqrt{11}+5 > \sqrt{11}+\sqrt{2}$

④ $2+\sqrt{7} > 4$

⑤ $-\sqrt{121} > -121$

6. $\boxed{1번}$

$\sqrt{5.83} = \sqrt{583.00} \times \sqrt{\frac{1}{10000}}$

$= 24.15 \times \frac{1}{100} = 2.415$

7. $\boxed{4번}$

$a=\sqrt{3}-2$ $b=\sqrt{3}+2$

$ab=(\sqrt{3}-2)(\sqrt{3}+2)$

$= -1$

a^2b^6+4a+1

$= a^2(ab)^6+4a+1$

$= (\sqrt{3}-2)^2+4(\sqrt{3}-2)+1$

$= 0.$

8. $\boxed{5번}$

$5<3+\sqrt{8}<6$

$9<5+\sqrt{17}<10$

① 자연수 자리 6.7.8.9.4개

② 무리수는 무한개.

③ 실수는 무한개

④ 가장 큰 정수는 9.

9. $\boxed{5번}$

$f(\frac{2}{3})=\sqrt{(\frac{3}{2})^2-1}=\sqrt{(\frac{3}{2}-1)\times(\frac{3}{2}+1)}$

$=\frac{\sqrt{5}}{2}$

$f(\frac{3}{5})=\sqrt{(\frac{5}{3})^2-1}=\sqrt{(\frac{5}{3}-1)\times(\frac{5}{3}+1)}$

$=\frac{\sqrt{16}}{3}=\frac{4}{3}$

$f(\frac{5}{7})=\sqrt{(\frac{7}{5})^2-1}=\sqrt{(\frac{7}{5}-1)\times(\frac{7}{5}+1)}$

$=\frac{\sqrt{24}}{5}=\frac{2\sqrt{6}}{5}$

$\frac{\sqrt{5}}{2}\times\frac{4}{3}\times\frac{2\sqrt{6}}{5}=\frac{4}{15}\sqrt{30}$

10. $\boxed{4번}$

$f(16)=2.$ $f(25)=2$

$f(36)=3.$ $f(49)=3.$

$f(20)+f(21)+\cdots+f(40)$

$=16\times2+3\times5=47$

11. $\boxed{3번}$

$ab^2>0$ $ab<0$

$a>0.$ $b<0$

$\sqrt{(ab)^2}+4\sqrt{(-ab)^2}$

$=|ab|+4|-ab|$

$=-ab-4ab$

$=-5ab.$

12. $\boxed{3번}$

$x^2-25x+10$ (×)

$(x-5)^2=x^2-10x+25$

13. $\boxed{2번}$

$\left(\frac{1}{2^2}-1\right)\left(\frac{1}{3^2}-1\right)\left(\frac{1}{4^2}-1\right)\cdots\left(\frac{1}{10^2}-1\right)$

$=\left(\frac{1}{2}-1\right)\left(\frac{1}{2}+1\right)\left(\frac{1}{3}+1\right)\left(\frac{1}{3}-1\right)\times\cdots\times\left(\frac{1}{10}-1\right)\left(\frac{1}{10}+1\right)$

$=-\frac{1}{2}\times\frac{3}{2}\times\frac{-2}{3}\times\frac{4}{3}\times\frac{-3}{4}\times\frac{5}{4}\times\cdots\times\frac{-9}{10}\times\frac{11}{10}$

$=-\frac{1}{2}\times\frac{11}{10}=-\frac{11}{20}$

14. $\boxed{5번}$

$(x+5)(x-A)$

$=x^2+(5-A)x-5A$

$-5A=15$

$A=-3$

$x^2+8x-15$

15. $\boxed{2번}$

$(x+5)(x-3)-k$

$=x^2+2x-15-k$

$-15-k=1$

$-k=16$

$k=-16$

16. $\boxed{1번}$

① 16 ② 2

③ 2 ④ 3 ⑤ 7

17. $\boxed{3번}$

$x^2-ax+21=(x-3)(x+b)$

$=x^2+(-3+b)x-3b$

$-3b=21$

$b=-7$ $a=10$

18. $\boxed{3번}$

$\sqrt{9a^2+54a+81} - 7\sqrt{\frac{16}{49} - \frac{16}{49}a + \frac{4}{49}a^2}$

$= \sqrt{9(a+3)^2} - 7\sqrt{\frac{4}{49}(a-2)^2}$

$= 3|a+3| - 7 \times \frac{2}{7}|a-2|$

$= 3(a+3) - 2(-a+2)$

$= 3a+9 + 2a - 4$

$= 5a+5.$

19. $\boxed{1번}$

$2x^2+8x+7x+28$

$= 2x^2+15x+28$

$(x+4)(2x+7)$

$(x+4+2x+7) \times 2$

$= 6x+22.$

20. $\boxed{5번}$

$(x+a)(x+b) = x^2+(a+b)x+ab$

$ab=1$ $(1,1)$ $1+1=2$ 소수.

$ab=4$ $(1,4)$ $1+4=5$ 소수

$(2,2)$ $2+2=4$. X

$ab=9$ $(1,9)$ $1+9=10$ x

$(3,3)$ $3+3=6$ x

$ab=16$ $(2,8)$ $2+8=10$ x

$(4,4)$ $4+4=8$ x

$ab=25$ $(5,5)$ $5+5=10$ x

$ab=36$ $(4,9)$ $4+9=13$ 소수

$ab=49$ $(7,7)$ $7+7=14$ x

$ab=64$ $(8,8)$ $8+8=16$ x

$ab=81$ $(9,9)$ $9+9=18$ x

$ab=100$ $(10,10)$ $10+10=20$ x

사실상 1.

(1). $(-3)^2=9$ 의 양의 제곱근

$\boxed{A=3}$

7의 음의 제곱근

$\boxed{B=-\sqrt{7}}$

(2). $\dfrac{1}{(3-\sqrt{7})} \times (3+\sqrt{7})$

$= \dfrac{3+\sqrt{7}}{9-7} = \boxed{\dfrac{3+\sqrt{7}}{2}}$

~~不合格~~ 2.

$a^2 - 9a + 14$

$= (a-2)(a-7)$

$= (7+\sqrt{3}-2)(7+\sqrt{3}-7)$

$= (5+\sqrt{3}) \times \sqrt{3}$

$= \boxed{5\sqrt{3}+3}$

~~不合格~~ 3.

$x^2 - 10mx + n - (2mx - 5n)$

$= x^2 - 12mx + 6n$.

$(-12m)^2 - 4 \times 6n = 0$.

$144m^2 = 24n$

$6m^2 = n$.

$m = 1 \quad n = 6$

$m = 2 \quad n = 24$

$m = 3 \quad n = 54$

$m = 4 \quad n = 96$

$n - m = 96 - 4 = \boxed{92}$

~~不合格~~ 4.

$\boxed{A = -4 - \sqrt{2^2+1^2} = -4-\sqrt{5}}$

$\boxed{B = -4 + \sqrt{5}}$

$\boxed{D = 2 + \sqrt{1^2+3^2} = 2+\sqrt{10}}$

$\boxed{C = 2 - \sqrt{1^2+3^2} = 2-\sqrt{10}.}$

$\overline{BC} = 2+\sqrt{10} - (-4+\sqrt{5})$

$= 6 + \sqrt{10} - \sqrt{5}$

$\overline{AD} = 2+\sqrt{10} - (-4-\sqrt{5})$

$= 6 + \sqrt{10} + \sqrt{5}$

$\overline{BC} - \overline{AD} = 6+\sqrt{10}-\sqrt{5}$

$-(6+\sqrt{10}+\sqrt{5})$.

$= \boxed{-2\sqrt{5}}$